Test Series No.

01

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

SYLLABUS

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A projectile launched on a horizontal surface (x-axis) follows a trajectory described by the equation $y = x - \frac{x^2}{20}$, where the y-axis is directed vertically upwards. The horizontal range of the projectile is: (all quantities are in SI units)

(1) 20 m (2) 10 m

(3) 40 m (4) 80 m

2. A heavy nucleus X with a mass number $A = 240$ has a binding energy per nucleon of 7.6 MeV. It undergoes fission and splits into two identical middle-mass fragments Y and Z, each with a mass number $A = 120$. If the binding energy per nucleon of the each fragment is 8.5 MeV, what is the total energy released (Q) in this nuclear fission reaction?

(1) 108 MeV (2) 216 MeV

(3) 24 MeV (4) 204 MeV

3. Consider a hypothetical hydrogen atom described by the Bohr model in which the mass of the electron is doubled. Let E_0 and r_0 represent the energy and radius of the first orbit (n = 1), respectively. Which of the following gives the correct values for E_0 and r_0? (Assume the standard Bohr radius is a_0 and the standard ground state energy is –13.6 eV)

(1) $E_0 = -27.2 \text{ eV}; r_0 = \frac{a_0}{2}$

(2) $E_0 = -27.2 \text{ eV}; r_0 = a_0$

(3) $E_0 = -13.6 \text{ eV}; r_0 = \frac{a_0}{2}$

(4) $E_0 = -13.6 \text{ eV}; r_0 = a_0$

4. When Young's double slit experiment is performed in vacuum, the 8th bright fringe is at distance 'X' from the central maxima. When the experiment is performed in a liquid, 9th dark fringe lies at same distance 'X' from the central maxima. What is the refractive index of the liquid?

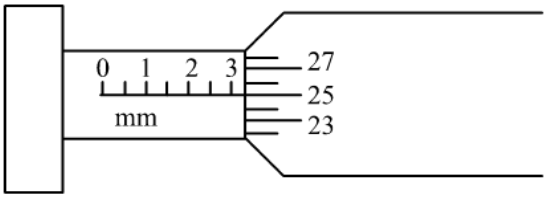
(1) 1.06

(2) 1.50

(3) 1.75

(4) 1.25

5. A screw gauge moves 1 mm on its linear scale for every 2 full rotations of its thimble, which contains 50 equal divisions. Before measurement, it was observed that when the studs are closed, the instrument has a negative zero error of 0.05 mm. If the instrument is used to measure the thickness of a metal plate as shown in the figure, the true thickness is:



(1) 3.30 mm (2) 3.20 mm

(3) 3.25 mm (4) 3.35 mm

6. The electric or magnetic field equations for four different electromagnetic waves propagating in various mediums (or vacuum) are given below. Which of the following equations represents a physically impossible electromagnetic wave? (Assume standard SI units for all constants).

(1) $E = E_0 \sin(1.5\times10^7 x - 4.5\times10^{15} t)$

(2) $B = B_0 \cos(2.0\times10^6 y + 4.0\times10^{14} t)$

(3) $E = E_0 \sin(1.0\times10^6 z - 5.0\times10^{14} t)$

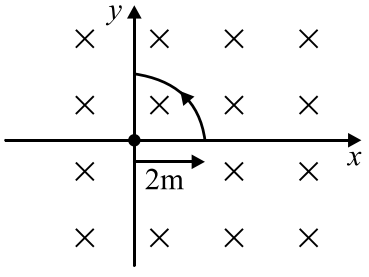
(4) $B = B_0 \cos(4.0\times10^7 z - 8.0\times10^{15} t)$

7. The equation of a particle performing SHM along x-axis about x = 0 is given by $x = 2\sin\left(2\pi t + \frac{\pi}{6}\right)$ (where x is in m and t is in s). What is the maximum acceleration of the particle?

(1) $8\pi^2$ m/s^2 (2) 8 m/s^2

(3) 4π m/s^2 (4) $4\pi^2$ m/s^2

8. A wire in form of a quarter circle is placed in the x-y plane with its center at the origin as shown. If the wire carries current $\sqrt{2}$A and there is a uniform magnetic field of 0.1 T present along Z-axis then what is the magnetic force acting on the wire?



(1) $\sqrt{2}\pi\times10^{-1}$ N (2) 40 N

(3) 0.4 N (4) 4 N

[Contd...

9. A small bar magnet is placed at the origin of a coordinate system such that its magnetic moment is along the positive x-axis. What is the direction of magnetic field due to the bar magnet at a point (– 10 cm, 0)?

(1) along $-x$ axis (2) along $+x$ axis

(3) along $+y$ axis (4) along $-y$ axis

10. Choose the option containing **correct** statements.

A. The minimum speed with which a car can move on a circular banked rough road of radius R, inclination θ and coefficient of friction μ is given by, $v_{min} = \sqrt{Rg\left(\frac{\tan\theta - \mu}{1 + \mu\tan\theta}\right)}$

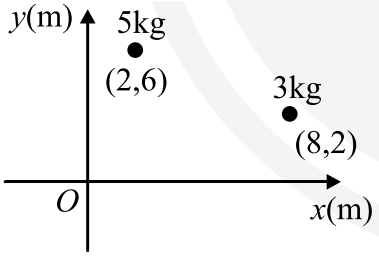
B. In case of vertical circular motion of a small ball using an inextensible string the minimum speed of the ball at the lowest point is $\sqrt{5}$ times of the minimum speed at the highest point while completing the vertical circle.

C. Variable spring force is conservative in nature.

(1) A and B only (2) A and C only

(3) B and C only (4) A, B and C

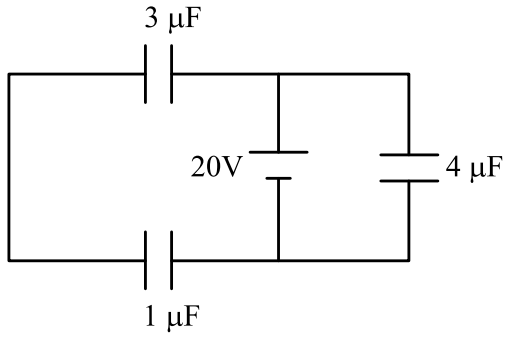
11. What is the distance of center of mass of the system of particles shown in figure from point $\left(\frac{1}{4}, \frac{3}{2}\right)$?



(1) 5 m (2) 3 m

(3) 4 m (4) 8 m

12. What is the sum of charges on the positive plates of 1μF and 4 μF capacitor in steady state?



(1) 80 μC (2) 95 μC

(3) 15 μC (4) 120 μC

13. The motion of a body in a straight line is given by the equation $\frac{dv}{dt} = 6t$, (where v is speed in m/s and t is time in second). If the body has velocity 2 m/s at $t = 0$, then the magnitude of velocity at $t = 1$ s is:

(1) 3 m/s

(2) 2 m/s

(3) 5 m/s

(4) 4 m/s

14. An aluminum wire and a brass wire of the same length are connected end-to-end. This composite wire is stretched by a force F as shown. If both wires have the same cross-sectional area, what can be said about the stress and strain in each wire? (Young's Modulus for brass is greater than for aluminum: $Y_{brass} > Y_{alum}$).



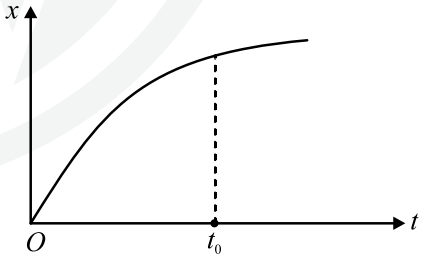
(1) Same stress, same strain.

(2) Same stress, different strain.

(3) Different stress, same strain.

(4) Different stress, different strain.

15. Position-time (x-t) graph of a particle moving in straight line is shown in figure. The average velocity of the particle from $t = 0$ to $t = t_0$ is v_{avg} and instantaneous velocity at time t_0 is v. Choose the **correct** option.



(1) $v > v_{avg}$

(2) $v < v_{avg}$

(3) $v = v_{avg}$

(4) Data insufficient

16. A block of mass 2 kg moves inside a frictionless circular track of radius 5m in a vertical plane. If speed of block at lowest point of track is 20 m/s, the normal force exerted on block by the track at lowest point is: (g = 10 m/s^2)

(1) 180 N

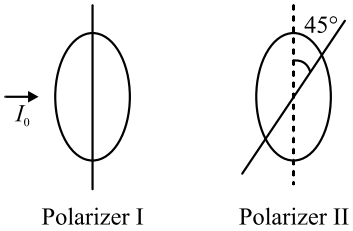
(2) 20 N

(3) 160 N

(4) 200 N

[Contd...

17. Unpolarized light with intensity I_0 is incident on a polaroid. Another polaroid having its transmission axis inclined at 45° to the first polarizer is also placed as shown. What is the ratio of intensity of light passing through first polarizer and intensity of light through the second polarizer?

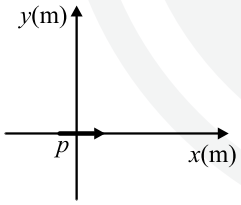


(1) 2 : 1 (2) 4 : 1
(3) 8 : 1 (4) 16 : 1

18. Kinetic energy of a particle moving along x-axis as a function of its position (r) and time (t) is given as $K = \frac{Ar^{2/3}}{B + t^3}$. If force acting on the particle depends on A, B and length L as $F = A^xB^yL^z$, then the value of $x^2 - y + z$ is:

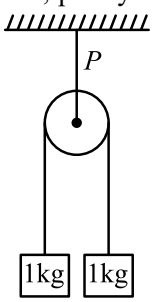
(1) $\frac{2}{3}$ (2) $\frac{5}{3}$
(3) $\frac{1}{3}$ (4) $\frac{4}{3}$

19. A short electric dipole having dipole moment p (in Cm) is placed on the origin along the x-axis as shown. What is the magnitude of net electric field due to the dipole at a point having coordinates (4m, 3m). (K is coulomb's constant in SI unit)



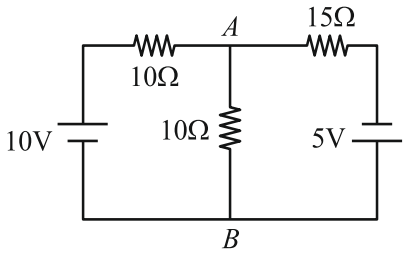
(1) $\frac{Kp\sqrt{73}}{625}$ (2) $\frac{Kp\sqrt{52}}{625}$
(3) $\frac{Kp\sqrt{43}}{625}$ (4) $\frac{Kp\sqrt{43}}{500}$

20. A wave pulse is generated in string P having linear mass density 2g/cm. What is the speed of the wave pulse? (g = 10 m/s^2, pulley is massless)



(1) $10\sqrt{2}$ m/s (2) 10 m/s
(3) 5 m/s (4) 20 m/s

21. Find the current in the branch AB in the given circuit

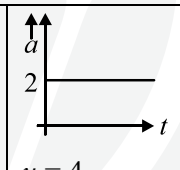
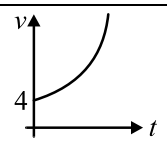
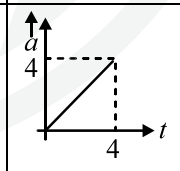
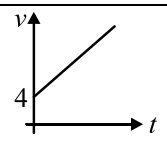
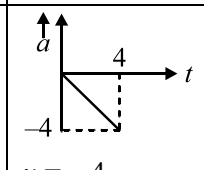
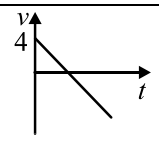
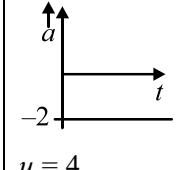
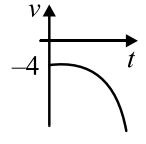


(1) 0.25 A (2) 0.50 A
(3) 0.125 A (4) 0.75 A

22. The magnetic susceptibility of a diamagnetic material depends on absolute temperature T as;

(1) $\chi \propto T$ (2) $\chi \propto \frac{1}{T}$
(3) $\chi \propto T^0$ (4) $\chi \propto \frac{1}{\sqrt{T}}$

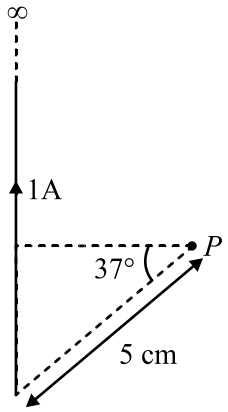
23. **List-I** contains acceleration time (a-t) graph of four particles while their corresponding velocity time (v-t) graphs are in **List-II**. Match the column and choose the **correct** option (symbols have usual meanings and physical quantities are in SI unit).

List-I		List-II	
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

[Contd...

24. A wire carries current of 1A as shown. What is the magnitude of magnetic field due to the wire at point *P*?



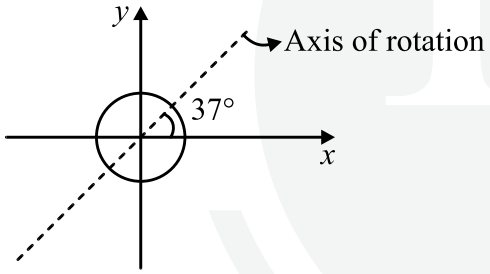
(1) 3.2×10^{-6} T

(2) 3.6×10^{-6} T

(3) 6.0×10^{-6} T

(4) 4.0×10^{-6} T

25. A uniform disc having mass *M* and radius *R* is placed in *xy* plane with its centre at the origin. What is the moment of inertia about the axis shown? (Axis of rotation is in the plane of the disc)



(1) $\frac{MR^2}{5}$

(2) $\frac{3MR^2}{4}$

(3) $\frac{MR^2}{2}$

(4) $\frac{MR^2}{4}$

26. A single circular loop of length *l* carrying current *I* has magnetic moment *M*. It is now bent to form a circular coil carrying *n* turns with same current. What is the magnetic moment of the new coil?

(1) $\frac{M}{n}$

(2) $\frac{M}{n^2}$

(3) n^2M

(4) nM

27. A circular coil is expanded radially in a region of magnetic field and no electromotive force is produced in the coil. Which of the following may be **correct**?

(1) There is a constant magnetic field in the perpendicular direction to the plane of coil.

(2) There is a decreasing magnetic field in the perpendicular direction to the plane of coil

(3) There is a increasing magnetic field in the perpendicular direction to the plane of coil

(4) Both (2) and (3)

28. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: On inserting a dielectric inside an initially charged isolated parallel plate capacitor, the energy stored by the capacitor increases.

Reason R: On insertion of dielectric, the capacitance of a parallel plate capacitor increases.

In the light of the above statements, choose the **correct** answer from the options given below:

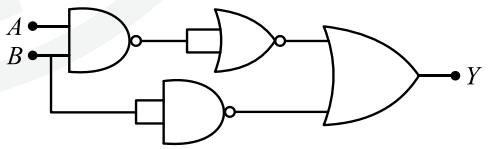
(1) A is true but R is false

(2) A is false but R is true

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

29. The output *Y* of the logic circuit shown in figure will be:



(1) $\overline{AB} + \overline{B}$ (2) $AB + \overline{B}$

(3) A (4) B

30. A metal rod 4 m long has a circular cross-section of radius 1 cm. One end of rod is kept at 100°C and the other at 20°C, then the temperature of cross-section at 50 cm from the hot end in steady state would be: (thermal conductivity of metal is 401 $Wm^{-1} K^{-1}$)

(1) 87.5°C

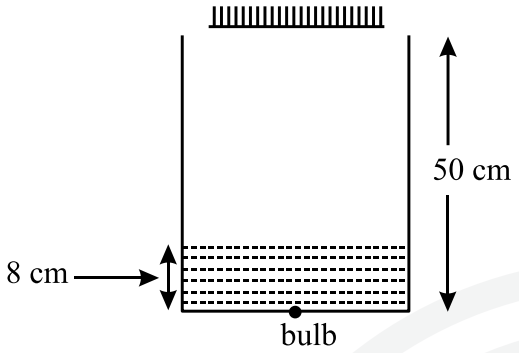
(2) 30°C

(3) 90°C

(4) 40°C

[Contd...

31. As shown in the figure, a plane mirror is fixed at a height of 50 cm from the bottom of tank containing water $\left[\mu = \frac{4}{3}\right]$. The height of water in the tank is 8 cm. A small bulb is placed at the bottom of the water tank. The distance of image of the bulb formed by mirror from the bottom of the tank is:

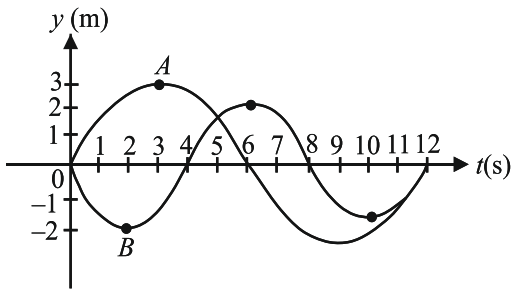


(1) 98 cm (2) 68 cm
(3) 72 cm (4) 88 cm

32. A river is flowing from west to east at a speed of 10 m/min. A man on north bank of river, capable of swimming at 20 m/min in still water wants to cross the river through the shortest path. In what direction should he swim?

(1) 30° West of north
(2) 30° North of west
(3) 30° West of south
(4) 30° South of west

33. The displacement *vs* time graph for two waves *A* and *B* which travel along the same string in two separate cases are shown in the figure. The ratio of their frequencies $\frac{f_B}{f_A}$ is;



(1) $\frac{9}{4}$ (2) $\frac{1}{1}$
(3) $\frac{81}{16}$ (4) $\frac{3}{2}$

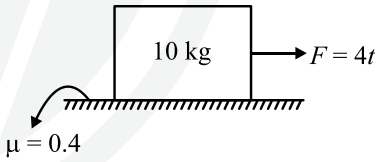
34. Consider a hollow uniformly charged, non-conducting sphere of radius *r*. If potential difference between its surface and at distance 3*r* from its centre is *V*, then magnitude of electric field due to it at distance 3*r* from its centre is:

(1) $\frac{V}{6r}$ (2) $\frac{V}{4r}$
(3) $\frac{V}{3r}$ (4) $\frac{V}{2r}$

35. Consider a closed circular coil of radius *r* having total number of turns *n* and total resistance *R*. If change of magnetic flux linked to each turn in given time duration is $\Delta\phi$ then total charge flown through it in that duration is:

(1) $\frac{\Delta\phi}{nR}$
(2) $\frac{n\Delta\phi}{R}$
(3) $\frac{R\Delta\phi}{n}$
(4) $\frac{r\Delta\phi}{n}$

36. A block of mass 10 kg is kept over a rough surface and a horizontal force $F = (4t)$ N is applied on it (where *t* is time in second). At what value of *t* will the block just start moving? ($g = 10$ m/s^2)



(1) 10 s (2) 8 s
(3) 9 s (4) 5 s

37. A ball at rest is released from a height equal to R_E above Earth's surface where R_E is the radius of Earth. What is the speed of the ball when it is at a distance $\frac{R_E}{2}$ above the Earth's surface? (M_E is mass of earth)

(1) $\sqrt{\frac{GM_E}{3R_E}}$
(2) $\sqrt{\frac{GM_E}{2R_E}}$
(3) $\sqrt{\frac{GM_E}{6R_E}}$
(4) $\sqrt{\frac{GM_E}{R_E}}$

[Contd...

38. The value of two resistors are $R_1 = (5 \pm 0.1)\text{k}\Omega$ and $R_2 = (20 \pm 0.4)\text{k}\Omega$. The maximum percentage error in the equivalent resistance when they are connected in parallel is:

(1) 5.125% (2) 2%

(3) 10.125% (4) 7%

39. Given below are two statements:

Statement I: In a series LCR a.c. circuit, the power factor at resonance is one.

Statement II: At resonance in a series LCR a.c. circuit, the voltage phasor and current phasor are perpendicular to each other.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

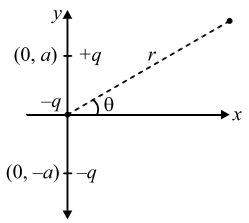
(4) Statement I and Statement II both are incorrect.

40. A stone is projected from the origin with an initial velocity u at an angle θ above the horizontal. What is the maximum value of θ such that the straight-line distance of the stone from the starting point always increases throughout its entire motion? (Neglecting air resistance)

(1) $\sin^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (2) $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{2}{3}\right)$

41. Three-point charges $-q$, $-q$ and $+q$ are situated in X-Y plane at points (0, $-a$), (0, 0) and (0, a) respectively. The potential at a distant point r ($r >> a$) in a direction making an angle θ with the X-axis is:



(1) $\frac{kq}{r}\left[1-\frac{2a\cos\theta}{r}\right]$ (2) $\frac{kq}{r}\left[1-\frac{2a\sin\theta}{r}\right]$

(3) $\frac{kq}{r}\left[\frac{2a\cos\theta}{r}-1\right]$ (4) $\frac{kq}{r}\left[\frac{2a\sin\theta}{r}-1\right]$

42. A particle starts rotating from rest about a fixed axis according to the formula $\theta = \left(\frac{3t^4}{8} - \frac{t^2}{2}\right)$ radians (where t is in seconds). The angular acceleration of the particle at the end of 2 seconds is;

(1) 10 rad/s^2 (2) 12 rad/s^2

(3) 16 rad/s^2 (4) 17 rad/s^2

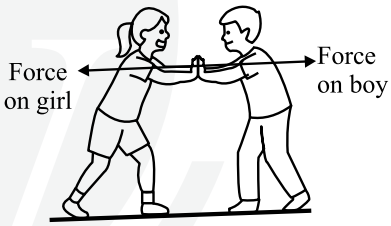
43. An electric lift with a maximum load of 2500 kg (lift + passengers) is moving up with a constant speed of 0.5 ms^{-1}. The frictional force opposing the motion is 3500 N. The minimum power delivered by the motor to the lift in watt is;

(g = 10 ms^{-2})

(1) 3000 W (2) 30000 W

(3) 28500 W (4) 14250 W

44. Two ice skaters, a man of mass 80 kg and a woman of mass 50 kg, are initially at rest on a flat ice bed. They push each other to gain some velocity. As a result, the man picks up a speed of 2.5 m/s. Consider the following statements:



Statement 1: The speed of the woman is 2.5 m s^{-1}.

Statement 2: The speed of the woman is 4.0 m s^{-1}.

Statement 3: Work done by the man while pushing the woman is 400 J.

Statement 4: Work done by the woman while pushing the man is less than the work done by the man pushing the woman.

Choose the **correct** options.

(1) Statements 1, 3 and 4 are only correct.

(2) Statements 1 and 4 only correct.

(3) Statements 2, 3 and 4 only correct.

(4) Statements 2 and 3 only correct.

45. The potential function of an electrostatic field is given by $V = 3x^2$ (where V is in volt and x is in metre). The electric field strength at the point (3m, 0, 4m) is;

(1) $-6\hat{i}$ N/C (2) $6\hat{i}$ N/C

(3) $18\hat{i}$ N/C (4) $-18\hat{i}$ N/C

[Contd...

46. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is:
(Given that 1 L bar = 100 J)
(1) –30 J (2) 5 kJ
(3) 25 J (4) 30 J

47. Which of the following has the highest negative electron gain enthalpy?
(1) Se (2) O
(3) Te (4) S

48. pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is:
(1) 0.5×10^{-15} (2) 0.25×10^{-10}
(3) 0.125×10^{-15} (4) 0.5×10^{-10}

49. Which of the following is **incorrect** statement about OSF_4?
(1) S atom has sp^3d^2 hybridization
(2) OSF_4 has trigonal bi pyramidal shape
(3) O atom is present at one of the equatorial position
(4) There is no lone pair on S

50. The increasing order of volatility of hydrides of group 16 elements is:
(1) $H_2O < H_2Te < H_2Se < H_2S$
(2) $H_2S < H_2O < H_2Se < H_2Te$
(3) $H_2O < H_2S < H_2Se < H_2Te$
(4) $H_2Te < H_2Se < H_2S < H_2O$

51. The aqueous solution having the highest freezing point is:
(1) 1 m glucose (2) 1 m KCl
(3) 1 m $Mg(NO_3)_2$ (4) 1 m $Al_2(SO_4)_3$

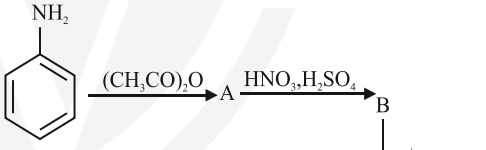
52. Given below are two statements:
Statement I: All B group vitamins can not be stored in our body.
Statement II: Lactose is a non-reducing sugar.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

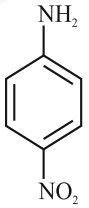
53. How many carbon atoms are present in 0.35 mole of $C_6H_{12}O_6$? (Given: $N_A = 6.023 \times 10^{23}$)
(1) 1.26×10^2 carbon atoms
(2) 1.26×10^{24} carbon atoms
(3) 1.26×10^{44} carbon atoms
(4) 1.26×10^{48} carbon atoms

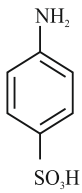
54. Number of Faraday required to obtain 2.5 mole of Ca from molten $CaCl_2$ is:
(1) 5F
(2) 2F
(3) 3F
(4) 4F

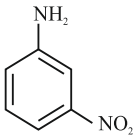
55. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is:
(Assume activation energy and pre-exponential factor are independent of temperature. ln(2) = 0.693; R = 8.314 J – mol^{-1} K^{-1})
(1) 100.6 kJ mol^{-1}
(2) 214.4 kJ mol^{-1}
(3) 107.2 kJ mol^{-1}
(4) 53.6 kJ mol^{-1}

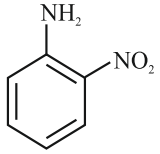
56. The major product 'P' obtained in the given reaction sequence is:



(1) 

(2) 

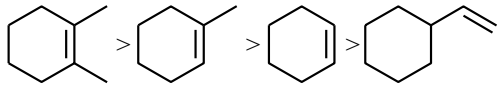
(3) 

(4) 

57. The **correct** order of the basic strength of methyl substituted amines in aqueous solution is:
(1) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$
(2) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$
(3) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$
(4) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

[Contd...

58. Conjugate base for Bronsted acids H_2O and HF are:
(1) OH^- and H_2F^+, respectively
(2) H_3O^+ and F^-, respectively
(3) OH^- and F^-, respectively
(4) H_3O^+ and H_2F^+, respectively

59. Which of the following is **incorrect** order?
(1) $CH_3^- > CH_3O^- > HO^- > H_2O$: (Nucleophilicity in protic solvent)
(2) $Cl^- > CH_3COO^- > CH_3O^- > NH_2^-$: (Leaving group ability)
(3)  : (Stability of alkene)
(4) $CH_3–CH_2–F > CH_3–CH_2–Cl > CH_3–CH_2–Br > CH_3–CH_2–I$: (Boiling point)

60. Consider the following reaction equilibrium
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
Initially, 1 mole of N_2 and 3 moles of H_2 are taken in a 2L flask. At equilibrium state if, the number of moles of N_2 is 0.6, what is the total number of moles of all gases present in the flask?
(1) 0.8 (2) 1.6
(3) 3.2 (4) 6.4

61. A compound (C_5H_8) react with ammonical $AgNO_3$ to give a white precipitate and reacts with excess of $KMnO_4$ solution to give $(CH_3)_2CH–COOH$. The compound is:
(1) $CH_2=CH–CH=CH–CH_3$
(2) $(CH_3)_2CHC\equiv CH$
(3) $CH_3(CH_2)_2C\equiv CH$
(4) $(CH_3)_2C=C=CH_2$

62. Fixed volume of 0.1 M benzoic acid solution is added into 0.2 M sodium benzoate solution and formed a 300 ml, resultant acidic buffer solution. If pH of this buffer solution is 4.5 then find added volume of benzoic acid (Given : pK_a benzoic acid = 4.2)
(1) 100 mL (2) 150 mL
(3) 200 mL (4) 250 mL

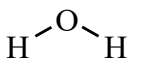
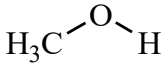
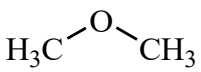
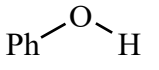
63. An electron moving with velocity 'v' is found to have a certain value of de-Broglie wavelength. The velocity to be possessed by the neutron to have the same de-Broglie wavelength is:
(1) $\frac{1840}{v}$ (2) $1840\ v$
(3) $\frac{v}{1840}$ (4) v

64. 2,2-Dichloropropane on hydrolysis yields:
(1) Acetone
(2) 2,2-Propane diol
(3) Isopropyl alcohol
(4) Acetaldehyde

65. Given below are two statements:
Statement I: $[Al(C_2O_4)_3]^{3-}$ is a homoleptic complex ion.
Statement II: Oxidation number of nickel in $[Ni(CO)_4]$ is zero.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

66. Dehydration of cyclopentyl carbinol with conc. H_2SO_4 forms:
(1) Cyclopentene
(2) Cyclohexene
(3) Cyclohexane
(4) Cyclopropene

67. Spin only magnetic moment of the complex ion, $[Ni(H_2O)_4]^{2+}$ is:
(1) zero BM
(2) $\sqrt{3}$ BM
(3) $\sqrt{24}$ BM
(4) $\sqrt{8}$ BM

68. The compound that is most difficult to protonate is:
(1) 
(2) 
(3) 
(4) 

69. The total volume of dry gaseous products at STP, when 3 moles of electrons are transferred from anode to cathode in the electrolysis of water is: (volume of gas at STP is 22.4 L)
(1) 67.2 L (2) 50.4 L
(3) 44.8 L (4) 56.0 L

[Contd...

70. Both acetaldehyde and acetone (individually) undergo which of the following reactions?

A. Iodoform Reaction
B. Cannizzaro Reaction
C. Aldol Condensation
D. Tollen's Test
E. Clemmensen Reduction

Choose the **correct** answer from the options given below:

(1) A, B and D Only
(2) A, C and E Only
(3) B, C and D Only
(4) C and E Only

71. Match the **List-I** with **List-II**.

List-I (Aquated metal ions)		List-II (Colour)	
(A)	Zn^{2+}	(I)	Blue
(B)	Mn^{2+}	(II)	Colourless
(C)	Cu^{2+}	(III)	Pink
(D)	Fe^{2+}	(IV)	Green

Choose the **correct** answer from the options given below;

(1) A-III; B-I; C-IV; D-II
(2) A-II; B-III; C-I; D-IV
(3) A-II; B-IV; C-I; D-III
(4) A-I; B-III; C-IV; D-II

72. What is the product when $C_6H_5–CH_2–NH_2$ reacts with nitrous acid?

(1) $C_6H_5-N\equiv N$ (2) $C_6H_5-CH_2-\overset{\oplus}{N}\equiv N$
(3) $C_6H_5-CH_2-OH$ (4) $C_6H_5-NH_2$

73. If the rate constant for a first order reaction is k, the time (t) required for the completion of 99% of the reaction is given by:

(1) t = 0.693/k (2) t = 6.909/k
(3) t = 4.606/k (4) t = 2.303/k

74. The structure and hybridisation of $Si(CH_3)_4$ are:

(1) bent, sp
(2) trigonal, sp^2
(3) octahedral, sp^3d^2
(4) tetrahedral, sp^3

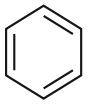
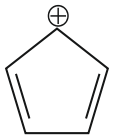
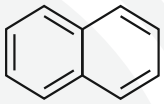
75. 0.15 g of hydrocarbon on complete combustion gives 0.44g of carbon dioxide. The percentages of carbon and hydrogen respectively are:

(1) 80 and 20 (2) 60 and 40
(3) 90 and 10 (4) 70 and 30

76. Which of the following conditions regarding a chemical process ensure its spontaneity at all temperature?

(1) $\Delta H > 0, \Delta S > 0$ (2) $\Delta H < 0, \Delta S > 0$
(3) $\Delta H < 0, \Delta S < 0$ (4) $\Delta H > 0, \Delta S < 0$

77. Which of the following is an example of antiaromatic species?

(1) 
(2) 
(3) 
(4) 

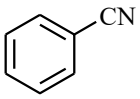
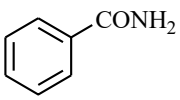
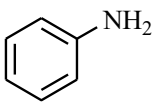
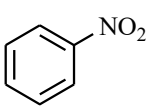
78. How many cyclic isomers of C_5H_{10} are possible? (excluding stereoisomer)

(1) 4 (2) 3
(3) 2 (4) 5

79. The major product obtain in reaction of t-butyl bromide with sodium methoxide is:

(1) t-butyl methyl ether
(2) Isobutylene
(3) Sodium t-butoxide
(4) Isobutane

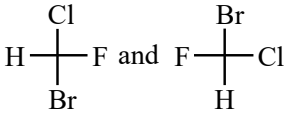
80. A given nitrogen-containing aromatic compound (A) reacts with Sn/HCl, followed by HNO_2 to give an unstable compound (B). (B), on treatment with phenol, forms a beautiful colour compound (C) with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound (A) is:

(1) 
(2) 
(3) 
(4) 

[Contd...

81. HCOOH on treating with conc. H_2SO_4 forms:
(1) CO_2
(2) CO
(3) Oxalic acid
(4) CH_3COOH

82. The following two compounds are:



(1) Enantiomers
(2) Diastereomers
(3) Identical
(4) Epimers

83. The enthalpy change for a given reaction at 298 K is –x J mol^{-1}. If the reaction occurs spontaneously at 298 K, the entropy change at that temperature:
(1) can be negative but numerically larger than x/298
(2) can be negative but numerically smaller than x/298
(3) cannot be negative
(4) cannot be positive

84. Which of the following statement is **true**?
(A) PH_5, NCl_5 and $BiCl_5$ do not exist
(B) I_3^+ has bent shape
(C) XeF_4 is a non polar molecule
(D) CO and C_2^{2-} do not have same bond order.
(1) A, B and D
(2) A, B and C
(3) B, C and D
(4) B and C only

85. Which of the following is **incorrect**?
(1) $CH_3-O-\overset{+}{C}H_2$ is more stable than $CH_3-\overset{+}{C}H_2$
(2) $(CH_3)_2\overset{+}{C}H$ is more stable than $CH_3-CH_2-\overset{+}{C}H_2$
(3) $CH_2=CH-\overset{+}{C}H_2$ is more stable than $CH_3-CH_2-\overset{+}{C}H_2$
(4) $CH_2=\overset{+}{C}H$ is more stable than $CH_3-\overset{+}{C}H_2$

86. If two substances A and B have $P_A^0 : P_B^0 = 1:2$ and have mole fraction in solution 1 : 2 then mole fraction of A in vapour phase will be:
(1) 0.33
(2) 0.25
(3) 0.20
(4) 0.52

87. If 'X' is ionization energy of hydrogen atom then the energy required for excitation of Li^{2+} ion electron from 2nd excited state to 5th excited state is:
(1) $\frac{3X}{4}$
(2) $\frac{4}{3X}$
(3) $\frac{X}{12}$
(4) $\frac{12}{X}$

88. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.2 M is 200 ohm . If the specific conductance is 0.4×10^{-3} S cm^{-1}, then the value of cell constant (in cm^{-1}) will be:
(1) 0.08 (2) 0.01
(3) 0.02 (4) 0.03

89. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Metallic conductance increases with increase of temperature.
Reason R: With increase in temperature vibration of metal atoms increases.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

90. Match the **List-I** with **List-II**.

List-I (Metal ion)		List-II (Group in qualitative analysis)	
(A)	Mn^{2+}	(I)	Group-III
(B)	As^{3+}	(II)	Group-II A
(C)	Cu^{2+}	(III)	Group-IV
(D)	Al^{3+}	(IV)	Group-II B

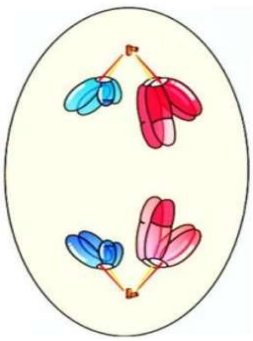
Choose the **correct** answer from the options given below;
(1) A-III; B-I, C-II; D-IV
(2) A-II; B-III, C-I; D-IV
(3) A-III; B-IV, C-II; D-I
(4) A-IV; B-II, C-I; D-III

[Contd...

91. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Green revolution was comparatively less effective in developing world where farmers were dependent on conventional breeding.
Reason: In developing world, inability to buy expensive agro-chemicals forced farmers to rely on conventional breeding.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

92. Given below are two statements:
Statement I: Phloem fibres are made up of collenchymatous cells.
Statement II: Phloem fibres of jute, flax and hemp are used commercially.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

93. Identify the **correct** stage of cell division shown in the following diagram.



(1) Prophase-I
(2) Metaphase-I
(3) Anaphase-I
(4) Telophase-I

94. In taxonomic hierarchy of animals, classes with a few similar characters are assigned to a higher category, called:
(1) phylum. (2) order.
(3) family. (4) division.

95. The number of recombinant therapeutics approved worldwide is ___**A**___ of which approx. ___**B**___ have been marketed in India.
Choose the **correct** option to fill the blanks.
(1) A: 30, B: 12
(2) A: 40, B: 20
(3) A:109, B: 32
(4) A: 111, B: 9

96. Find the **incorrect** match.
(1) 1983- Eli Lilly created human insulin
(2) 1990- First gene therapy
(3) 1997- First transgenic cow Rosie
(4) 1994- American company got patent on Basmati rice

97. Identify the number of chromosomes in G_1 and G_2 phases when a cell is dividing, if this diploid cell initially has 48 chromosomes.
(1) 24 and 48 respectively
(2) 6 and 24 respectively
(3) 12 and 24 respectively
(4) 48 and 48 respectively

98. Nucleolus found in a cell is:
(1) single membrane bound.
(2) always one in number per nucleus.
(3) structure present in cytoplasm.
(4) the site for active ribosomal RNA synthesis.

99. Activated sludge in a sewage treatment is:
(1) grit in sewage.
(2) large masses of anaerobic bacteria.
(3) settled bacterial flocs.
(4) mixture of gases.

100. *Equisetum* plant belongs to the class:
(1) Pteropsida. (2) Sphenopsida.
(3) Psilopsida. (4) Lycopsida.

101. Which of the following features are commonly used as the basis for the classification of animals?
(1) Arrangement of cells, body symmetry, and nature of coelom
(2) Color of the body, size of the animal, and habitat
(3) Shape of the head, length of the tail, and body weight
(4) Type of locomotion, presence of feathers, and diet

[Contd...

102. Which of the following is the respiratory organ of land insects?
(1) Tracheal tubes (2) Gills
(3) Cuticle (4) Lungs

103. How many O_2 molecules can bind with a single molecule of Hb?
(1) 8
(2) 6
(3) 4
(4) 5

104. Choose the statement which is **incorrect** for embryo development in flowering plant.
(1) Occurs at micropyle end of ovule.
(2) Early stages of embryo development differs in monocots and dicots.
(3) Globular embryo stage preceed heart shaped embryo stage.
(4) Most zygote divide only after certain amount of endosperm is formed.

105. What is the primary role of the descending limb of Henle's loop?
(1) To reabsorb electrolytes actively.
(2) To concentrate the filtrate by being permeable to water.
(3) To reabsorb HCO_3^- and secrete hydrogen ions.
(4) To maintain high osmolarity of medullary interstitial fluid.

106. Match **List-I** with **List-II**:

List-I		List-II	
(A)	*Physalia*	(I)	Sea-fan
(B)	*Adamsia*	(II)	Portuguese man-of-war
(C)	*Pennatula*	(III)	Sea anemone
(D)	*Gorgonia*	(IV)	Sea-pen

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-IV, B-I, C-II, D-III

107. Which one of the following is **not** correctly matched?
(1) Dengue : *Aedes* mosquito
(2) Malaria : Female *Anopheles*
(3) Amoebiasis : Houseflies
(4) Ringworm : Droplets

108. Extract of _____ and _____ have hallucinogenic properties.
(1) A- *Cannabis sativa*, B- *Papaver somniferum*
(2) A- Heroin, B- *Cannabis sativa*
(3) A- *Erythroxylum coca*, B- *Papaver somniferum*
(4) A- *Atropa belladona*, B- *Datura*

109. Read the following statement and select the **incorrect** ones:
A. Na^+/K^+ ATPase pump helps to maintain an electrochemical ionic gradient across the axolemma.
B. At rest, axoplasm has lower Na^+ ion concentration than K^+ ion concentration.
C. The endocrine system provides chemical integration through hormones.
D. Thalamus, a part of the hindbrain, is responsible for emotions like anger and rage.
E. Transmission across chemical synapse is slower than along electrical synapse.
(1) D only (2) D and E
(3) B and D (4) A and C

110. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Ringworm	(I)	*Haemophilus influenzae*
(B)	Filariasis	(II)	*Trichophyton*
(C)	Malaria	(III)	*Wuchereria bancrofti*
(D)	Pneumonia	(IV)	*Plasmodium vivax*

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-II, C-I, D-IV

111. Which of the following is seen in renal calculi?
(1) Deposition of sand particles in urethra.
(2) Deposition of protein in urethra.
(3) Crystallisation of oxalates in kidney.
(4) Deposition of fat in artery.

112. Identify the **correct** features of Class Aves:
A. Presence of feathers and beak.
B. Forelimbs are modified into wings.
C. Long bones are hollow with air cavities (pneumatic).
D. Air sacs connected to lungs supplement respiration.
E. They are poikilothermous (cold-blooded).
(1) A, B, C and D only
(2) A, C, D and E only
(3) B, C and E only
(4) All are correct

[Contd...

113. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Urine formed by the nephrons is ultimately carried to the urinary bladder.
Reason R: Urinary bladder stores the urine till a involuntary signal is given by the central nervous system (CNS).
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

114. Identify the **incorrect** statement for dense connective tissues.
A. The cells and fibres are loosely arranged in semi-fluid ground substance.
B. Tendons and ligaments are included in dense connective tissues.
C. Orientation of fibres show a regular or irregular pattern in dense connective tissues.
D. Fibres and fibroblasts are compactly packed in dense connective tissues.
Choose the **correct** option:
(1) Only A (2) A and B
(3) B and C (4) A, C and D

115. Which of the following is **correct** about viroids?
(1) They have free RNA without protein coat.
(2) They have DNA with protein coat.
(3) They have free DNA without protein coat.
(4) They have RNA with protein coat.

116. Select the **incorrect** pair from the following.
(1) Amphibians of plant kingdom - Bryophytes
(2) Sexual reproduction accompanied by complex post fertilisation developments - Rhodophyceae
(3) Water required for fertilization - Pteridophytes
(4) First terrestrial plants to possess vascular tissues - Gymnosperms

117. Which among the following is **not** a common feature of wind pollinated flowers?
(1) Pollen grains are light and non-sticky
(2) Large often-feathery stigma
(3) Numerous flowers packed into an inflorescence
(4) Colourful and produce nectar

118. Consider the following statements:
A. Somatic neural system relays impulses from CNS to skeletal muscles.
B. Autonomic neural system transmits impulses from CNS to involuntary organs and smooth muscles.
C. Unmyelinated nerve fibres are commonly found in spinal and cranial nerves.
Which of the above statements are **true**?
(1) A and B only
(2) A and C only
(3) B and C only
(4) A, B, and C

119. Identify the **incorrect** statement from the following regarding sex determination in human beings.
(1) Sex determining mechanism in case of humans is XX type.
(2) Out of 23 pairs of chromosomes present, 22 pairs are autosomes.
(3) A pair of X-chromosomes are present in the female.
(4) Presence of an X and Y chromosome are determinant of the male characteristic.

120. Chief producers in the oceans are:
(1) euglenoids.
(2) dinoflagellates.
(3) diatoms.
(4) slime moulds.

121. Out of the following cranial meninges, which one is found in close proximity with brain tissue?
(1) Arachnoid (2) Pia mater
(3) Dura mater (4) Both (1) and (3)

122. A typical nucleosome in eukaryotes contains:
(1) 200 bp of DNA helix.
(2) 100 bases of DNA helix.
(3) 200 bases of DNA helix.
(4) 100 bp of DNA helix.

123. Select the **correct** statements regarding epidermis.
A. Epidermal tissue system forms the outer-most covering of the whole plant body.
B. It is made up of elongated cells.
C. It is always multi-layered.
D. Epidermal cells have a large vacuole.
E. Epidermal cells are parenchymatous.
Choose the *most appropriate* answer from the options given below:
(1) B, C and D only
(2) A, B, C and D only
(3) A, B, D and E only
(4) A and B only

124. A piece of DNA is subjected to restriction enzyme digestion and the digested sample is put under electrophoresis. Which statement is **correct** in above context?
(1) Smallest fragment is present near cathode
(2) Smallest fragment is in the middle of gel
(3) Largest fragment is in the middle of gel
(4) Smallest fragment is present near anode

125. Which of the following statement is **incorrect**?
(1) Smoking reduces the concentration of haembound oxygen.
(2) Nicotine raises blood pressure and increases heart rate.
(3) Smoking is associated with increased incidence of cancer.
(4) Smoking causes oxygen deficiency only in lungs.

126. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against the pO_2.
Reason R: Oxygen dissociation curve is highly useful in studying the effect of factors like pCO_2, H^+ concentration, etc., on binding of O_2 with haemoglobin.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

127. Select the **correct** pair from the following.
(1) Poisonous insects- Consumed in large quantities by predators
(2) Frogs - Use cryptic coloration to get detected by predators
(3) Monarch butterfly- Distasteful to birds due to chemicals acquired during the caterpillar stage
(4) Predators - Prudent behaviour overexploits prey

128. Termination in polypeptide chain during translation process is brought about by codons:
(1) UUG, UAG and UAA.
(2) UUU, UGA and AUG.
(3) UAG, UGA and UCA.
(4) UAA, UAG and UGA.

129. The enzyme DNA ligase is used:
(1) for template independent polypeptide formation.
(2) to start DNA replication.
(3) for identification of promoter on dsDNA.
(4) to join the discontinuously synthesised DNA fragments during replication.

130. Given below are two statements:
Statement I: Yeasts poison themselves to death only when the concentration of alcohol reaches about 18 per cent.
Statement II: Pyruvic acid dehydrogenase and alcohol decarboxylase catalyse the reactions of fermentation in yeast.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

131. The common feature between *Psittacula* and *Pteropus* is:
(1) mammary glands.
(2) homeothermy.
(3) viviparity.
(4) pneumatic bones.

[Contd...

132. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Recombinant DNA	(I)	Sea weeds
(B)	Cellulase	(II)	DNA staining
(C)	Ethidium bromide	(III)	Isolation of DNA from plant cell
(D)	Agarose	(IV)	Artificial DNA molecule

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

133. What is the utility of the bacterium, *Thermus aquaticus*?
(1) It is used in disarming Ti DNA.
(2) It is used to create recombinant plasmids only.
(3) It is used in autoradiography.
(4) It facilitates the polymerase chain reaction.

134. Match **List-I** with **List-II** w.r.t a *lac* operon.

List-I		List-II	
(A)	Operator site	(I)	Encode proteins required for lactose metabolism
(B)	Promoter site	(II)	Repressor binding site
(C)	Structural gene	(III)	RNA polymerase binding site
(D)	Regulator gene	(IV)	Produces repressor protein

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-II, B-III, C-I, D-IV

135. Identify the statements that are **correct** regarding cytokinins.
A. Helps in formation of adventitious shoots.
B. Help to overcome the apical dominance.
C. Speeds up the malting process in brewing industry.
D. Induce flowering in mango.
E. They were first discovered as kinetin.
Choose the *most appropriate* answer from the options given below:
(1) A, B and C only
(2) A, B and E only
(3) B, D and C only
(4) B, C and E only

136. Match **List-I** with **List-II:**

List-I		List-II	
(A)	Tendons	(I)	Protect neurons
(B)	Fibroblasts	(II)	Attach bones to bones
(C)	Neuroglial cells	(III)	Attach skeletal muscles to bones
(D)	Ligaments	(IV)	Areolar connective tissue

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-II, C-I, D-III

137. Which of the following statements is **incorrect**?
(1) *Trichoderma* are free-living fungi, that are very common in root ecosystems of plants.
(2) *Aspergillus niger* produces butyric acid.
(3) The 'Roquefort cheese' are ripened by growing a specific fungi on them.
(4) *Oscillatoria* is autotrophic microbe that helps in nitrogen fixation.

138. Choose the **correct** statement with respect to G_0 phase.
(1) Cells that do not divide further exit G_2 phase to enter G_0 phase.
(2) Cells of this stage remain metabolically inactive and no longer proliferate.
(3) Cells of this stage remain metabolically active but no longer proliferate.
(4) Cells of the epidermis in the adult animals do not appear to exhibit division and are present in this stage.

139. The plant body of liverworts:
(1) is sporophyte.
(2) is thalloid.
(3) possesses roots.
(4) has frond.

140. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Use of chitinase enzyme is necessary for isolation of DNA from yeast cells but not in case of bacteria.
Reason R: Fungal cell wall is made up of fungal cellulose or chitin.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

[Contd...

141. In *Calotropis* flower aestivation is:
(1) valvate.
(2) vexillary.
(3) imbricate.
(4) twisted.

142. A person suffering from a disease caused by *Plasmodium*, experiences recurring chill and fever at the time when;
(1) the microgametocytes and mega gametocytes are being destroyed by the WBCs.
(2) the sporozoites released from RBCs are being rapidly killed and broken down inside spleen.
(3) the *Plasmodium* reach maximum growth and give out certain toxins.
(4) the parasite after its rapid multiplication inside RBCs ruptures them, releasing the stage to enter fresh RBCs.

143. Which of the following best describes similarity between DNA and RNA?
(1) Both are always double-stranded.
(2) Both have same sugar molecules.
(3) Both have same pyrimidine bases.
(4) Both are polymers of nucleotides.

144. Which one of the following statements regarding auxins is **correct**?
(1) IAA is natural auxin but NAA, 2,4-D are synthetic auxins.
(2) NAA is natural auxin but IBA and 2,4-D are synthetic auxins.
(3) 2,4-D is natural auxin but IAA and IBA are synthetic auxins.
(4) IAA, NAA, IBA and 2, 4-D are synthetic auxins.

145. The fruit in Cruciferae and Malvaceae is called as:
(1) siliqua only.
(2) capsule only.
(3) siliqua and capsule respectively.
(4) drupe only.

146. What is the reason for death of bollworms due to Bt toxin protein:
(1) Inhibition of nerve impulse.
(2) Disturbed digestion of food in foregut.
(3) Creation of pores that cause cell swelling and lysis.
(4) Lysis of cells of hindgut.

147. Which of the following sets includes only the bacterial disease?
(1) Cholera, typhoid, smallpox.
(2) Tetanus, dengue and chikungunya.
(3) Malaria, smallpox and poliomyelitis.
(4) Diphtheria, dysentery and plague.

148. **X** is the right granted by a government to an inventor to prevent others from commercial use of his invention. When '**X**' is granted for biological entities and for products derived from them, these are called **Y**.
Read the above paragraph and identify **X** and **Y**.
(1) X - patent, Y - biopatent
(2) X - piracy, Y – biopiracy
(3) X - patent, Y - biopiracy
(4) X - piracy, Y –biopatent

149. $\log S = \log C + Z\log A$
In the given equation of species-area relationship, the value of regression coefficient for a whole continent, would be:
(1) 0.1 – 0.2 (2) 0.5 – 0.6
(3) 0.6 – 1.2 (4) 0.3 – 0.5

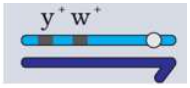
150. Refer to the statements **(A-C)** given below and identify the type of tissue.
A. They are found in the inner linings of hollow organs such as fallopian tubes and bronchioles.
B. Their function is to move particles or mucus in a specific direction over it.
C. The shape of cells can be cuboidal or it can be columnar also.
(1) Glandular epithelium
(2) Compound epithelium
(3) Ciliated epithelium
(4) Squamous epithelium

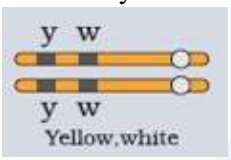
151. Identify from the following which of the following feature belongs to members of kingdom Monera.
(1) They lack cell wall.
(2) All are heterotrophic.
(3) Though the bacterial structure is very simple, they are very complex in behaviour.
(4) Membranous inclusion bodies.

152. If a recombinant DNA is inserted within the coding sequence of β-galactosidase enzyme, then;
(1) recombinant colonies will give blue colour in presence of chromogenic substrate.
(2) non-recombinant colonies will produce blue colour in presence of chromogenic substrate.
(3) non-recombinant colonies will not produce colour due to insertional inactivation.
(4) both recombinant and non-recombinant colours will produce blue colour.

[Contd...

153. Haemophilia will occur in 50% daughters when:
(1) When father is haemophilic and mother is normal.
(2) When father is normal and mother is haemophilic .
(3) When father is haemophilic and mother is carrier.
(4) When father and mother both are haemophilic.

154. Read the following and select the **incorrect** statement.
(1) 

represents dominant wild type genotype in *Drosophila*.
(2) A cross between plants differing in two traits is called dihybrid cross.
(3) 

represents male with recessive linked genes in *Drosophila*.
(4) Chromosomes as well as genes occur in pairs.

155. Select the **incorrect** statements.
A. Plant families Convolvulaceae and Solanaceae are included in the order Polymoniales.
B. Dog and cat belong to the same family.
C. Lion and panther belongs to same phylum primata.
D. Species is the lowest category.
Choose the *most appropriate* answer from the options given below:
(1) B and C only
(2) A, B and C only
(3) A, B and D only
(4) A, C and D only

156. Passive immunity is defined as immunity;
(1) inherited from parents.
(2) achieved through first exposure to the disease.
(3) achieved through vaccination.
(4) injecting ready made antibodies directly.

157. Which of the following conditions is/are associated with emphysema?
A. It is an acute disorder.
B. Alveolar walls are damaged.
C. Wheezing sound is produced.
D. Respiratory surface is increased.
(1) A and B only
(2) B only
(3) C and D only
(4) D only

158. Which of the following is a completely curative method for treatment of ADA deficiency in a patient?
(1) Functional ADA is given to the patient by injection.
(2) Infusion of lymphocytes containing functional ADA cDNA after birth.
(3) Gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages.
(4) Bone marrow transplantation in which ADA gene is introduced into adult cells only.

159. Which of the following options is **correct** for the matrix of cartilage?
(1) It is soft and pliable.
(2) It is hard and non-pliable.
(3) It is solid and pliable.
(4) It is soft and non-pliable.

160. Match **List-I** with **List-II**:

List-I		**List-II**	
(A)	Atrial Natriuretic factor	(I)	Adrenal cortex
(B)	Renin	(II)	Vasodilation
(C)	Angiotensin II	(III)	Vasoconstrictor
(D)	Aldosterone	(IV)	Juxtaglomerular cells

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-I, C-II, D-III

161. In the World Summit (held in Johannesburg)____**A**____ countries pledged their commitment to achieve by ____**B**____a significant reduction in the current rate of biodiversity loss at global, regional and local level. Identify **A** and **B** respectively.
(1) A-190, B-2012
(2) A-192, B-2002
(3) A-190, B-1992
(4) A-190, B-2010

162. Read the following statements related to HIV/AIDS and select the **correct** option:

A. HIV uses the enzyme reverse transcriptase to convert its RNA genome into DNA inside host cells.

B. Prevention is the best option for AIDS because currently, there is no permanent cure for the disease.

C. AIDS is a contagious disease that spreads through the air like the common cold.

D. The main goal of WHO's HIV/AIDS prevention programs is to promote education, awareness, and preventive measures.

(1) Only A and C are correct

(2) Only B, C, and D are correct

(3) Only A, B, and D are correct

(4) A, B, C, and D are correct

163. Bt toxin genes have been expressed in plants in order to provide resistance against;

(i) lepidopterans and fungi.

(ii) animals and bacteria.

(iii) bacteria and fungi.

(iv) coleopterans and dipterans.

(v) lepidopterans.

(1) (ii) and (iii) only

(2) (i) and (iv) only

(3) (iii) and (v) only

(4) (iv) and (v) only

164. The point of attachment of funicle with the body of the ovule in a typical anatropous ovule is:

(1) nucellus. (2) hilum.

(3) micropyle. (4) chalaza.

165. Given below are two statements:

Statement I: Cytokinesis is similar in plant and animal cells.

Statement II: In plants the cell division takes place via formation of cell furrow.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

166. Which of the given sets include the primary lymphoid organs?

(1) Thymus, lymph nodes and spleen.

(2) Bone marrow and thymus.

(3) Bone marrow, Peyer's patches and thymus.

(4) Thymus, liver and tonsils.

167. Residual volume is:

(1) additional volume of air, a person can inspire by a forcible inspiration.

(2) total volume of air accommodated in the lungs at the end of a forced inspiration.

(3) volume of air remaining in the lungs even after inspiration.

(4) volume of air remaining in the lungs even after a forcible expiration.

168. Given below are two statements:

Statement I: There is a linear relationship between incident light and CO_2 fixation rates at low light intensities.

Statement II: Light saturation for photosynthesis occurs at 20 per cent of the full sunlight.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

169. Following are the side effect of excessive use of anabolic steroid **except**;

(1) enlargement of clitoris in females.

(2) reduction in size of testicles in males.

(3) mood swings in both females and males.

(4) reduced aggression in both male and female.

170. Which of the following statements is **true** regarding the digestive system in Platyhelminthes?

(1) It has a complete digestive system with separate mouth and anus.

(2) It has an incomplete digestive system with a single opening.

(3) It has no digestive system at all.

(4) It has a closed circulatory system associated with digestion.

[Contd...

171. For glycolysis to continue, cell will require:
(1) O_2 and mitochondria.
(2) mitochondria and citrate synthase.
(3) acetyl CoA and pyruvate decarboxylase.
(4) NAD^+ and hexokinase.

172. Given below are two statements:
Statement I: During urine formation, the tubular cells secrete substances like H^+, K^+ and ammonia into the filtrate.
Statement II: Tubular secretion is the first step in urine formation.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

173. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Metacentric chromosome	(I)	Centromere situated close to the end forming one extremely short and one very long arms
(B)	Acrocentric chromosome	(II)	Centromere at the terminal end
(C)	Sub-metacentric chromosome	(III)	Centromere in the middle forming two equal arms of chromosomes
(D)	Telocentric chromosome	(IV)	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the **correct** answer from the options given below:
(1) A-III B-IV C-II D-I
(2) A-II B-III C-IV D-I
(3) A-I B-III C-IV D-II
(4) A-III B-I C-IV D-II

174. Among the following, the stroma lamellae membranes have:
(1) PS I only.
(2) PSI and PSII both, and enzymes for Calvin cycle.
(3) PS II and NADP reductase.
(4) PS I, enzymes for C_4 pathway and NADP reductase.

175. The annual net primary productivity of land is approximately about :
(1) 170 billion tons of (dry weight) of organic matter.
(2) 70 billion tons of (dry weight) of organic matter.
(3) 55 billion tons of (dry weight) of organic matter.
(4) 115 billion tons of (dry weight) of organic matter.

176. Select **correct** statements for dorsiventral leaves.
(1) They are present in monocots.
(2) They are present in dicots.
(3) They have parallel venation.
(4) They have near similar sizes of vascular bundles.

177. Which step from the following defines the action of citrate synthase enzyme?
(1) Succinate → Oxaloacetate
(2) Isocitrate → α - ketoglutarate
(3) Citrate → Isocitrate
(4) Oxaloacetate + Acetyl CoA → Citric acid

178. From the given list, identify the number of plants with superior ovary.
Cucumber, ray florets of sunflower, rose, plum, guava, peach and china rose
Choose the *most appropriate* answer from the options given below:
(1) One (2) Three
(3) Four (4) Five

179. The process of mineralisation by micro-organisms helps in the release of:
(1) inorganic nutrients from humus.
(2) both organic and inorganic nutrients from detritus.
(3) organic nutrients from humus.
(4) inorganic nutrients from detritus and formation of humus.

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions.
Reason R: Medulla is a part of forebrain present between thalamus and hypothalamus.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

SPACE FOR ROUGH WORK



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if